National Institute of Diabetes and Digestive and Kidney Diseases

Focal Neuropathies

In this section:

What are focal neuropathies?

Focal neuropathies are conditions in which you typically have damage to single nerves, most often in your hand, head, torso, or leg. This type of nerve damage is less common than peripheral or autonomic neuropathy.

Many different focal neuropathies can affect people who have diabetes.

Entrapments

Entrapments, or entrapment syndromes, are the most common type of focal neuropathy. Entrapments occur when nerves become compressed or trapped in areas where nerves pass through narrow passages between bones and tissues. People with diabetes are more likely to have entrapments than people without diabetes.[1]

The most common entrapment is called carpal tunnel syndrome NIH. Although less than 10 percent of people with diabetes feel symptoms of carpal tunnel syndrome, about 25 percent of people with diabetes have some nerve compression at the wrist.[2]

Other focal neuropathies

Other focal neuropathies that do not involve trapped nerves are much less common. These focal neuropathies most often affect older adults. Examples include cranial neuropathies, which affect the nerves of the head. Cranial neuropathies can cause eye problems or problems with the muscles of the face. Symptoms depend on which nerve is affected.

What causes focal neuropathies?

Over time, high blood glucose, also called blood sugar, and high levels of fats, such as triglycerides, in the blood from diabetes can damage your nerves and the small blood vessels that nourish your nerves, leading to focal neuropathies.

What are the symptoms of focal neuropathies?

Entrapments—focal neuropathies that involve trapped nerves—cause symptoms that begin gradually and get worse over time. Examples include

- carpal tunnel syndrome, which causes pain, numbness, and tingling in your thumb, index finger, and middle finger, and sometimes weakness of your grip
- ulnar entrapment, which causes pain, numbness, and tingling in your little and ring fingers
- peroneal entrapment, which causes pain on the outside of your lower leg and weakness in your big toe

The most common entrapment is carpal tunnel syndrome.

Focal neuropathies that do not involve trapped nerves cause symptoms that begin suddenly and improve after several weeks or months. Depending on which nerve is affected, you may have pain and other symptoms in your

- hand
- leg
- foot
- torso

Cranial neuropathies—focal neuropathies that affect the nerves in the head—may cause symptoms such as

- aching behind one eye
- double vision
- paralysis on one side of your face, called Bell's palsy
- problems focusing your eyes

How do doctors diagnose focal neuropathies?

Doctors diagnose focal neuropathies by asking about your symptoms and performing tests, such as nerve conduction studies NIH and electromyography (EMG) NIH. Nerve conduction studies check how fast electrical signals move through your nerves in different parts of your body. EMG shows how your muscles respond to your nerves.

How do doctors treat focal neuropathies?

Your doctor may treat focal neuropathy pain with the same medicines used to treat peripheral neuropathy pain.

To treat a focal neuropathy that involves a trapped nerve, your doctor may recommend

- wearing a splint or brace to take pressure off the nerve
- medicines that reduce inflammation
- surgery, if other treatments don't work

For focal neuropathies that don't involve trapped nerves, most people recover within a few weeks or months, even without treatment.

References

[1] Pop-Busui R, Boulton AJ, Feldman EL, et al. Diabetic neuropathy: a position statement by the American Diabetes Association. *Diabetes Care.* 2017;40(1):136–154.

[2] Izenberg A, Perkins BA, Bril V. Diabetic neuropathies. *Seminars in Neurology.* 2015;35(4):424–430.

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Previous: Autonomic Neuropathy

Next: Proximal Neuropathy

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